

PB-MFS: Multi-Funnel Selection with Pareto-Balanced Uncertainty and Diversity for Bioacoustic Active Learning

Haoyu Wang^{1,2,*}, Han Dou^{3,*}, Hui Chen³, Yushan Du¹, Liushi Tu³,
Ningning Pan¹, Hao Zhang³, Jingyao Wu⁴, Gang Li², Gongping Huang^{3,†}

¹ Southwestern University of Finance and Economics, China

² MiLM Plus, Xiaomi Inc., China

³ Electronic Information School, Wuhan University, China

⁴ Massachusetts Institute of Technology, USA

Abstract—Active learning mitigates annotation costs in bioacoustic classification; however, effective batch selection remains challenging for multi-label audio data. Conventional uncertainty-based strategies frequently sample redundant instances, whereas diversity-based approaches tend to overlook informative samples crucial for optimizing the current classifier. To address these limitations, we propose Pareto-Balanced Multi-Funnel Selection (PB-MFS), a novel batch active learning strategy that decouples candidate filtering from final batch selection. Specifically, PB-MFS initially constructs a survivor set via multiple complementary acquisition funnels. Subsequently, it determines the optimal annotation batch by harmonizing uncertainty and diversity through a Pareto optimization objective. The proposed method integrates diversity-driven cold-start selection, multi-funnel survivor voting, and an uncertainty-diversity trade-off leveraging pretrained audio embeddings. Comprehensive evaluations on the four BioDCASE 2026 Task 4 subsets (HSN, POW, UHH, and ATBFL) demonstrate that PB-MFS achieves an aggregated mean AULC of 0.510. Furthermore, it outperforms the strongest baseline, CoreSet, by an absolute margin of 0.050, delivering a 10.9% relative performance gain.

Index Terms—active learning, bioacoustics, multi-label classification, uncertainty sampling, diversity sampling

1. INTRODUCTION

Bioacoustic monitoring has become a vital tool for monitoring biodiversity, ecosystem dynamics, and species activities. Although modern recording devices can collect large volumes of audio with minimal human intervention, obtaining reliable annotations remains expensive and time-consuming [1]–[3]. This challenge is particularly evident in multi-label bioacoustic classification, where a single audio segment often contains overlapping acoustic events from multiple sources. Therefore, reducing annotation effort while maintaining model performance is critical for practical bioacoustic learning systems.

Active learning addresses this challenge by iteratively selecting small batches of unlabeled samples for annotation and retraining the model with the expanded labeled dataset [4]. Its central goal is to identify the most informative samples under a limited annotation budget. In batch active learning, the problem becomes more difficult because the selected samples should not only be informative individually but also provide complementary information as a group [5].

In multi-label bioacoustic classification, a single acquisition criterion is often insufficient. Uncertainty sampling methods tend to select redundant samples [6], whereas diversity-based methods may overlook informative ones [7]. Recent work combines uncertainty and diversity to improve batch selection [8], [9]. Following this direction by integrating multiple acquisition signals before the final selection stage, we propose Pareto-Balanced Multi-Funnel Selection (PB-MFS), a

batch active learning method for multi-label bioacoustic classification. PB-MFS separates candidate filtering from final batch selection. It first constructs a compact survivor set using multiple acquisition criteria and then selects the final annotation batch with a Pareto-based objective that balances uncertainty and diversity.

Our contributions are as follows:

- 1) We introduce a multi-funnel candidate filtering strategy that combines complementary acquisition criteria.
- 2) We propose a Pareto-based batch selection objective for uncertainty-diversity balancing.
- 3) Experiments on the BaseAL benchmark show that PB-MFS consistently outperforms strong baselines under limited annotation budgets.

2. RELATED WORK

Active learning reduces annotation costs by selecting informative unlabeled samples under limited annotation budgets. Settles [4] provides a comprehensive overview of active learning paradigms, which are designed to reduce annotation effort. This framework is particularly suitable for bioacoustic monitoring, where passive recordings require extensive expert annotation for species identification and multi-label event recognition. Recent applications in environmental audio [10], sound event detection [11], and wildlife acoustic monitoring [12] demonstrate the importance of efficient sample selection under limited resources.

Existing active learning methods are commonly categorized into uncertainty-based, diversity-based, and representative approaches. Uncertainty sampling [6], including least-confidence and margin-based strategies, prioritizes ambiguous samples near the decision boundary. Diversity-based methods, such as diversity-aware batch selection [13], CoreSet [7], and farthest-first traversal, improve batch coverage by reducing redundancy among selected samples. Representative methods, such as TypiClust [14], combine clustering with typicality to capture common patterns in the data while providing limited coverage for rare events.

Hybrid active learning methods combine multiple acquisition signals to overcome the limitations of individual criteria. In multi-label settings, previous studies integrate uncertainty with representativeness or diversity to improve sample selection [15]–[17]. BADGE [8] jointly models uncertainty and diversity through gradient embeddings, while VAAL [18] learns an acquisition strategy in latent space. These approaches are relevant to bioacoustic learning, where datasets often contain sparse events, imbalanced classes, and complex environmental noise. Rauch et al. [19] investigate annotation scarcity and domain variation in avian bioacoustics, while Zhang and Virtanen [20] combine committee disagreement with farthest-first traversal for sound event detection.

*Equal contribution. †Corresponding author: gongpinghuang@whu.edu.cn.

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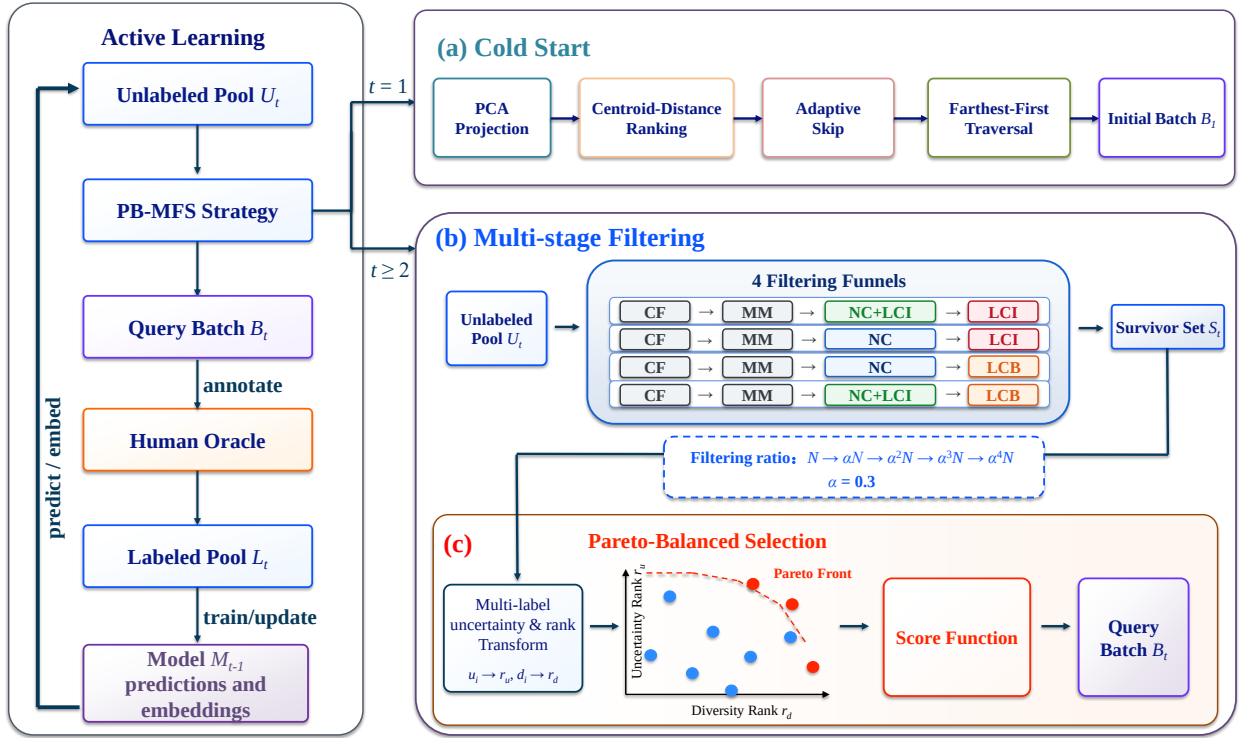


Fig. 1: Overview of the Pareto-Balanced Multi-Funnel Selection (PB-MFS) framework. At $t = 1$, PCA-based farthest-first sampling constructs a diverse initial batch. For $t \geq 2$, four ordered filtering funnels generate survivor lists that are merged by voting. The final stage forms the query batch from uncertainty and PCA-64 diversity ranks.

Although hybrid strategies have shown promising results, balancing multiple acquisition objectives remains largely dependent on heuristic rules. Pareto-based optimization provides a principled mechanism for representing trade-offs among competing objectives [21]. In this work, PB-MFS introduces a Pareto-based selection framework that jointly optimizes uncertainty and diversity for multi-label bioacoustic classification.

3. PROPOSED METHOD

3.1. Problem Formulation

In each active learning cycle t , the learner maintains a labeled set $\mathcal{L}_t$ and an unlabeled pool $\mathcal{U}_t$, with $\mathcal{L}_t \cap \mathcal{U}_t = \emptyset$. Each audio segment i is represented by a fixed embedding z_i extracted from a pretrained audio model. For every $i \in \mathcal{L}_t$, the corresponding multi-label target is $y_i = [y_{i1}, \dots, y_{iC}]^T \in \{0, 1\}^C$, where C denotes the number of classes; this follows the standard multi-label classification setting [22]. Given an annotation budget k_t , the learner selects a query batch $\mathcal{B}_t \subseteq \mathcal{U}_t$ with $|\mathcal{B}_t| = k_t$. After that, the queried samples are added to the labeled set and removed from the unlabeled pool:

$$\mathcal{L}_{t+1} = \mathcal{L}_t \cup \mathcal{B}_t, \quad \mathcal{U}_{t+1} = \mathcal{U}_t \setminus \mathcal{B}_t, \quad (1)$$

where $\setminus$ denotes set difference, indicating that the queried batch $\mathcal{B}_t$ is removed from the unlabeled pool $\mathcal{U}_t$.

At the beginning of cycle $t \geq 2$, the classifier $f_{\theta_{t-1}}$ trained after the previous acquisition is available and produces predictions $p_i = f_{\theta_{t-1}}(z_i) \in [0, 1]^C$ for each candidate in $\mathcal{U}_t$. The ideal query batch would maximize the expected performance of the retrained model on unseen data:

$$\mathcal{B}_t^* = \arg \max_{\substack{\mathcal{B} \subseteq \mathcal{U}_t \\ |\mathcal{B}|=k_t}} \mathbb{E}_{Y_{\mathcal{B}} | \mathcal{L}_t, \mathcal{U}_t} [\mathcal{M}(f_{\theta_t}(\mathcal{B}, Y_{\mathcal{B}}))], \quad (2)$$

where $Y_{\mathcal{B}}$ denotes the unknown labels of the candidate batch and $\mathcal{M}$ is the evaluation metric. Since these labels are unavailable at acquisition time, PB-MFS approximates this objective using the available embeddings and model predictions.

3.2. PB-MFS Framework

Figure 1 illustrates the overall framework of PB-MFS. The method consists of three modules. During the first active learning cycle, Diversity-Based Cold Start selects $\mathcal{B}_1$ directly from the embedding space because prediction scores are not yet available. In later cycles, Multi-Funnel Candidate Filtering takes the labeled set, the unlabeled pool, the embeddings, and the model predictions as input to produce a survivor set $\mathcal{S}_t$. Pareto-Balanced Batch Selection then constructs the final query batch $\mathcal{B}_t$ from $\mathcal{S}_t$ by balancing uncertainty and diversity. This design separates efficient candidate screening from the more expensive batch selection, allowing the filtering stage to preserve candidates supported by different acquisition signals while the final stage identifies a balanced and informative annotation batch.

3.3. Acquisition Strategy

3.3.1. Diversity-Based Cold Start: When $t = 1$, the labeled set is empty and uncertainty estimates are not available. PB-MFS performs farthest-first traversal in a PCA-reduced embedding space to construct the initial batch [23]. The PCA projection preserves the overall geometric structure while reducing the computational cost of repeated distance calculations. Standard farthest-first traversal tends to select isolated outliers as initial seeds. To mitigate this, PB-MFS first ranks candidates according to their distance from the embedding centroid and excludes part of the extreme tail. The initial seed is then chosen from a restricted range using the largest local distance gap, and the remaining samples are selected greedily by repeatedly choosing the point farthest from the current selected set. This procedure ensures

broad coverage of the embedding space before acquisition becomes available.

3.3.2. Multi-Funnel Candidate Filtering: For later cycles, applying the final selection directly to the entire unlabeled pool is computationally expensive and does not fully exploit complementary information. PB-MFS first constructs a candidate pool $\mathcal{P}_t \subseteq \mathcal{U}_t$ and passes it through a series of filtering funnels. Within each funnel, every criterion retains the highest-ranked candidates from the output of the previous stage. Thus, the filtering process progressively removes less informative candidates using embedding geometry, prediction uncertainty and label structure.

Five acquisition criteria are used: CoreSet Farthest (CF), Multi-label Margin (MM), Label Class Balance (LCB), Neighborhood Consistency (NC), and Label Cardinality Inconsistency (LCI). CF measures the distance to the labeled set, MM estimates prediction uncertainty, LCB prioritizes underrepresented classes, NC compares predictions with local label statistics inspired by ML-KNN [24], and LCI measures the deviation of predicted label cardinality from the labeled-set average. Their scores are computed as:

$$q_{\text{CF}}(i) = \min_{j \in \mathcal{L}_t} \|z_i - z_j\|_2^2, \quad (3)$$

$$q_{\text{MM}}(i) = 1 - \frac{1}{0.5C} \sum_{c=1}^C |p_{ic} - 0.5|, \quad (4)$$

$$q_{\text{LCB}}(i) = \sum_{c=1}^C \frac{p_{ic}}{n_c + 1}, \quad (5)$$

$$q_{\text{NC}}(i) = \frac{1}{C} \sum_{c=1}^C |p_{ic} - \bar{y}_{ic}^{(K)}|, \quad (6)$$

$$q_{\text{LCI}}(i) = \left| \sum_{c=1}^C p_{ic} - \frac{1}{|\mathcal{L}_t|} \sum_{j \in \mathcal{L}_t} \sum_{c=1}^C y_{jc} \right|, \quad (7)$$

where $q_{\text{CF}}(i)$, $q_{\text{MM}}(i)$, $q_{\text{LCB}}(i)$, $q_{\text{NC}}(i)$, and $q_{\text{LCI}}(i)$ denote the five corresponding scores of candidate i . Moreover, n_c is the number of labeled samples in class c , and $\bar{y}_{ic}^{(K)}$ is the mean ground-truth label for class c among the K nearest labeled neighbors of candidate i . Intuitively, CF promotes exploration of uncovered regions, MM favors class-wise probabilities close to 0.5, LCB prioritizes underrepresented classes, NC captures local prediction-label inconsistency, and LCI quantifies the deviation between predicted and average labeled-set cardinality.

Each filtering step reduces a working set of size N to $\max(4k_t, \text{round}(\alpha N))$ candidates. PB-MFS adopts four funnels that share a CF-MM prefix—first covering unlabeled regions, then prioritizing uncertain predictions—and then branch into complementary tails, as shown in Fig. 1: NC+LCI→LCI, NC→LCI, NC→LCB, and NC+LCI→LCB. These combinations mix neighborhood inconsistency, cardinality mismatch, and class-balance cues rather than committing to a single heuristic. For each candidate, we compute its retention count across funnels:

$$v_i = \sum_f \mathbf{1}\{i \in \mathcal{S}_t^{(f)}\}. \quad (8)$$

Candidates with higher retention counts are retained first, ensuring consensus across multiple acquisition paths. The final survivor set $\mathcal{S}_t$ contains at least $\max(4k_t, 100)$ samples and is passed to the final selection stage.

3.3.3. Pareto-Balanced Batch Selection: The final stage selects k_t samples from $\mathcal{S}_t$. To combine uncertainty and diversity without relying on their original numerical scales, both criteria are converted

into normalized ranks. We define a normalized multi-label uncertainty score as:

$$u_i = \frac{1}{C} \sum_{c=1}^C 4p_{ic}(1 - p_{ic}). \quad (9)$$

After applying an adjustment $\tilde{u}_i = \max(\epsilon, u_i)^\gamma$, we map $\tilde{u}_i$ to a min-max normalized rank $r_u(i) \in [0, 1]$, where $r_u(i) = 1$ denotes the most uncertain survivor in $\mathcal{S}_t$.

For diversity estimation, embeddings in $\mathcal{S}_t$ are projected and normalized. Let $\mathcal{B}_t^{(m)}$ denote the partial batch after m selections. The first sample is the candidate with the highest uncertainty rank. For each subsequent step $m \geq 2$, the diversity score of an unselected candidate is

$$d_i = \min_{j \in \mathcal{B}_t^{(m-1)}} \|z_i - z_j\|_2^2, \quad (10)$$

and these scores are converted into min-max normalized diversity ranks $r_d(i) \in [0, 1]$, where $r_d(i) = 1$ denotes the candidate farthest from the current partial batch. The next sample is chosen by maximizing

$$a(i) = \min(r_u(i), r_d(i)) + \eta \sqrt{r_u(i)r_d(i)}. \quad (11)$$

The first term penalizes candidates with a low rank in either criterion, while the second term favors samples that perform well on both. Consequently, the selected batch avoids highly uncertain but redundant samples as well as diverse but uninformative ones. Diversity ranks are updated after each selection, allowing subsequent decisions to adapt to the current batch composition.

4. EXPERIMENTS AND RESULTS

4.1. Experimental Setup

We evaluate PB-MFS on the BioDCASE 2026 Task 4 active learning benchmark [25]. Our local evaluation covers four subsets: the terrestrial BirdSet subsets HSN, POW, and UHH [26], and the marine ATBFL subset [27]. All four subsets are represented by standardized BaseAL bundles containing `perch_v2` embeddings [28]. All methods follow a common active learning protocol. We allocate a total labeling budget of $M = 500$ samples to each subset over 16 acquisition cycles, with batch sizes following a fixed schedule. After each acquisition cycle, we train the embedding classifier for 10 epochs using mini-batches of 32 samples. All methods use the same fixed embeddings, classifier architecture, and training pipeline so that performance differences primarily reflect the acquisition strategy. Each configuration is run five times independently.

For PB-MFS, cold-start selection uses a 256-dimensional PCA projection [30]. The filtering stage operates on the original acoustic embeddings, whereas the final diversity ranking uses 64-dimensional PCA embeddings. We set $\alpha = 0.3$ and use $K = 5$ labeled neighbors for computing NC. Candidates receiving at least three votes are prioritized when constructing $\mathcal{S}_t$. For final batch selection, we set $\epsilon = 0.05$, $\gamma = 1.0$, and $\eta = 0.25$.

We compare PB-MFS with the official BaseAL implementations of Random, Margin, CoreSet, and TypiClust. Performance is measured using the area under the learning curve (AULC), defined as the normalized area under the macro mean average precision (mAP) curve:

$$\text{AULC} = \frac{1}{M} \sum_{t=1}^T (n_t - n_{t-1}) \text{mAP}_{\text{macro}}(n_t), \quad (12)$$

where n_t denotes the cumulative number of labeled samples after cycle t , and $n_0 = 0$. Wall-time is the average CPU sampling time per acquisition step (excluding classifier retraining). Annotation cost

Table 1: Baseline comparison under the BaseAL protocol. Best results are in **bold**; second-best are underlined. Arrows indicate whether higher ($\uparrow$) or lower ($\downarrow$) is better.

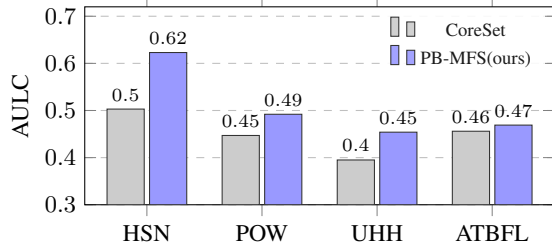
Method	AULC (macro mAP) $\uparrow$	Wall-time (s) $\downarrow$	Ann. Cost $\downarrow$
Random	0.390	0.00131	966.85
Margin	0.399	<u>0.00307</u>	1267.75
CoreSet	<u>0.460</u>	5.04364	1019.05
TypiClust	0.423	5.90751	958.10
PB-MFS(ours)	0.510	1.15090	1146.55

follows the BaseAL protocol and accumulates a per-sample cost that depends on the predicted label cardinality of queried instances, so methods that select denser multi-label clips incur a higher cost under the same sample budget.

4.2. Local Evaluation Results

Table 1 summarizes the local evaluation results across the four subsets. PB-MFS achieves an AULC of 0.510, compared with 0.460 for CoreSet, the strongest baseline. This corresponds to an absolute improvement of 0.050 under the same labeling budget. The average sampling time of PB-MFS is 1.15090 seconds per acquisition step. It is faster than CoreSet and TypiClust, although slower than Random and Margin. These results indicate that PB-MFS achieves the best AULC while maintaining a moderate sampling cost.

Figure 2 compares the performance of PB-MFS and CoreSet on each subset. PB-MFS achieves a higher AULC across all four subsets. The largest improvement is observed on HSN, where the AULC increases from 0.503 to 0.623. On POW, UHH, and ATBFL, the corresponding improvements are 0.045, 0.059, and 0.013, respectively. These results show that PB-MFS consistently outperforms CoreSet on both the terrestrial and marine datasets.

**Fig. 2:** AULC comparison between CoreSet and PB-MFS across four BioDCASE 2026 Task 4 subsets under the BaseAL protocol.

4.3. Official Held-out Evaluation

Table 2 presents the reserved evaluation results provided by the BioDCASE 2026 Task 4 organizers. PB-MFS achieves an AULC of 0.499 with an average sampling time of 0.485 seconds per acquisition step, ranking third on the final leaderboard. ADU-MMR [29] achieves the highest AULC of 0.507, exceeding PB-MFS by 0.008, but requires 1.006 seconds per sampling step. In comparison, PB-MFS uses less than half of the sampling time reported for ADU-MMR on the held-out evaluation. Among the baseline methods, Random and Margin are fastest but have the lowest AULC, whereas TypiClust and CoreSet achieve higher AULC at substantially longer sampling times. PB-MFS offers a favorable balance between AULC and sampling efficiency.

4.4. Ablation Results and Hyperparameter Sensitivity

Table 3 summarizes the ablation study. Removing the cold start causes the largest drop (AULC 0.510 $\rightarrow$ 0.483), confirming that a diverse seed set strongly shapes later cycles under a 500-sample budget. Using one

Table 2: Official held-out evaluation results for BioDCASE 2026 Task 4.

Method	AULC $\uparrow$	Sampling Time (s) $\downarrow$	Ann. Cost $\downarrow$
Random	0.401	0.001	966.050
Margin	0.408	<u>0.002</u>	1263.650
TypiClust	0.421	5.076	958.950
CoreSet	0.464	5.838	1023.600
PB-MFS(ours)	<u>0.499</u>	0.485	1109.900
ADU-MMR	0.507	1.006	1026.700

funnel instead of four reduces AULC by only 0.005 while lowering wall-time from 0.8561 s to 0.6265 s; removing the Pareto selector yields a 0.002 drop. These smaller late-stage gains are expected: after a strong cold start, the remaining pool is already partially covered, so complementary filtering and rank balancing mainly reduce redundancy and stabilize selection rather than rewrite the learning curve. They remain useful for accuracy–efficiency trade-offs across domains, as reflected by consistent subset-level gains in Figure 2.

Table 3: Ablation study on PB-MFS components.

Method	AULC $\uparrow$	Wall-time (s) $\downarrow$	Ann. Cost $\downarrow$
PB-MFS	0.510	0.8561	<u>1145.60</u>
PB-MFS w/ One Funnel Chain	0.505	<u>0.6265</u>	1141.50
PB-MFS w/o Cold Start	0.483	0.5957	1163.20
PB-MFS w/o Pareto	<u>0.508</u>	0.7801	1150.15

Table 4 shows the sensitivity analysis for the PCA dimension, retention ratio, and balance coefficient. Among the tested values, the default setting achieves the highest AULC in each group. Increasing the retention ratio from 0.3 to 0.5 decreases the AULC from 0.510 to 0.496, representing the largest performance change among the evaluated settings. For the PCA dimension, the 64- and 512-dimensional configurations both achieve an AULC of 0.508, while the 128-dimensional configuration reaches 0.501. Varying η from 0.0 to 0.5 results in AULC values between 0.506 and 0.507. In general, the retention ratio has the greatest influence on performance among the three hyperparameters.

Table 4: Hyperparameter sensitivity of PB-MFS. Default settings are marked with an asterisk (*).

Hyperparameter	Value	AULC $\uparrow$	Δ AULC
Cold-Start PCA Dim.	64	0.508	-0.002
Cold-Start PCA Dim.	128	0.501	-0.009
Cold-Start PCA Dim.	256*	0.510	–
Cold-Start PCA Dim.	512	<u>0.508</u>	-0.002
Retention Ratio α	0.2	<u>0.503</u>	-0.007
Retention Ratio α	0.3*	0.510	–
Retention Ratio α	0.5	0.496	-0.014
Balance Coef. η	0.0	0.506	-0.004
Balance Coef. η	0.25*	0.510	–
Balance Coef. η	0.5	<u>0.507</u>	-0.003

5. CONCLUSION AND LIMITATIONS

In this paper, we present PB-MFS, a multi-stage active learning method for bioacoustic classification under limited annotation budgets. PB-MFS combines a diversity-based cold start, multi-funnel survivor voting, and Pareto-based batch selection that balances uncertainty and diversity. Experiments show competitive cross-domain performance and third place in the official evaluation. The method still depends on pretrained audio embeddings and several hyperparameters; future work will simplify the framework and evaluate it beyond BaseAL and BioDCASE Task 4.

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